Advanced Engineered Software, Inc.

Bitcode

Supply & Demand — The Two Sides of the Market

A live, measured market for code IP. Depositors turn repositories into measured AssetPacks; buyers acquire exactly the IP their needs require — priced in the open, settled in a couple of clicks, and minted into $BTD.

Field	Detail
Purpose	Define the two-sided market: supply, demand, differentiation vs today's solutions, Bitcode's niche, the full buyer/seller universe, and the transactions Bitcode is built for
Core unit	AssetPack (AP) — a synthesized, measured patchfile of code IP
Core asset	$BTD — a deflationary token minted on purchase, backed by measured quantity of knowledge
Companion	Reads with the Bitcode Market Analysis — Research Brief (sizing & accounts)
Date	July 17, 2026

Internal strategy document. Market figures and competitor facts are drawn from third-party sources and are directional; product mechanics reflect Bitcode's current design.

Contents

1. Executive summary

Bitcode is a two-sided market for code IP. On the **supply** side, an IP-holder connects a repository, marks obfuscations (what to withhold), and Bitcode synthesizes a measured **AssetPack** — a patchfile scored by exhaustive static analysis and AI evaluation. On the **demand** side, a buyer states a need; Bitcode synthesizes an AssetPack from the relevant deposited IP and attaches *dynamic*, need-relative measurements showing exactly how well it will satisfy that need before a cent is spent.

Two properties make Bitcode a market rather than a catalog. First, **measurement**: every pack carries an objective quality scores (measurements) and a *quantity-of-knowledge* scalar ($BTD), so IP can be priced and compared. Second, **liquidity**: purchases transfer unlimited rights (resale, hoarding, open-sourcing) and mint **$BTD** — a deflationary, knowledge-backed token — turning code IP into a tradeable asset class. Prices are set by live bidding, in the open.

The one-line thesis. Today's data channels are slow (licensing deals run ~9 months and often collapse; dead-code IP takes ~11 months and clears at ~5% of estimate), opaque (closed-door, NDA pricing), and open only to a handful of large sellers. Bitcode makes code IP **fast, transparently priced, and liquid for everyone** — and is the only venue that measures quality, discovers price live, and clears rights for resale, natively for code.

What this document covers

- How today's solutions work and where they break (§2) — the friction Bitcode removes.
- The Bitcode model (§3) — AssetPacks, measurement, $BTD, rights, and pricing.
- Supply (§4) and demand (§5) — the mechanics, the economics, and the full universe of sellers and buyers.
- Differentiation and niche (§6–§7) — faster, live, easier; and the white space Bitcode occupies.
- Real transactions (§8) — the concrete deals Bitcode is more optimal for than any broker.

2. How today's solutions work — and where they break

Four channels move data/IP to AI builders today. Each solves one piece and leaves the expensive pieces — price discovery, fit measurement, rights, and liquidity — unsolved.

2.1 Bespoke licensing (the publisher-deal model)

Buyers negotiate directly with large rights-holders. It works for a handful of premium sellers and no one else. Deals take **months** and frequently fail (OpenAI–NYT ran ~9 months before the Times walked and sued; AFP–Mistral came only “after months of in-depth discussions”). Pricing is confidential and “disguised” inside broad partnerships, and “rates vary by orders of magnitude.” As of mid-2025 only ~52 licensing deals were publicly known — 38 sellers to 12 buyers, in just 5 languages. Individual holders earn “a few dollars”; the long tail has no bargaining power. ***Press Gazette; Quartz; Neudata (2024–26).***

2.2 Managed data vendors (Scale, Surge, Mercor)

These firms **manufacture** data with contracted labor rather than trade existing IP — “specialized human-made data” produced to spec. Contracts are bespoke and opaque ($100k–$400k+ annually, “no public pricing schedule”), and relationships, not prices, move volume (Google, at ~$200M/yr, severed Scale after Meta's stake). There is no inventory of pre-existing IP, no open price, and no resale. ***Marketplace/APM; Sacra; Teahose (2025–26).***

2.3 Data marketplaces (AWS Data Exchange, Snowflake, Databricks, Hugging Face)

Listing venues that solve “search” and little else. Together they capture **<2% of data-trade volume**; there is no real-time price discovery (“a closed-door negotiation place”), no measurement of whether a dataset satisfies a buyer's specific need, and rights are frequently unclear — 65% of Hugging Face dataset licenses are omitted or mislabeled, and 10–20% default silently to “all rights reserved.” Platform incentives favor their own compute, not the trade. ***Data Boutique (2023); arXiv 2502.04484; truelabel.ai (2025–26).***

2.4 Tokenized data & IP attempts

A crypto-native cohort has tried to make data/IP tradeable, and splits cleanly into two camps that never meet. A **measurement** camp (Numerai, Bittensor) scores contributions with staking and burns — but for predictions and model outputs, not tradeable IP, and inside closed systems. A **provenance** camp (Story Protocol → “DATA Foundation,” Ocean, Vana, Sahara, Grass) tokenizes data/IP with on-chain attribution — but does *not* measure quality or run live per-asset bidding, and often encumbers rights (royalties, non-transferable shares). Pure tokenized-IP exchanges have failed outright: IPXI ceased in 2015 and IPwe went bankrupt in 2024 — both on illiquidity, weak demand, and no enforcement. ***CoinDesk; Messari; Contreras (2016); IAM (2024–26).***

2.5 Dead-code & IP liquidation

When a company dies, its code is monetized — if at all — as patents via bankruptcy auctions. The process is slow and buyer-dominated, and there is no price discovery: Kodak's imaging patents, valued up to $4.5B, sold for **$94M (~4%)** after an ~11-month process, while Nortel's raised 450% of estimate — the same asset class swinging from 5% to 450% because no live market sets a clearing price. “In most cases, founders are exhausted and just walk away,” and the code “usually ends up forgotten.” ***IEEE Spectrum (2014); TechCrunch; FasterCapital.***

2.6 The friction, summarized

Pain point	How it shows up today	Evidence
Latency	Months per bespoke deal; no real-time path	OpenAI–NYT ~9 mo (failed); dead-IP ~11 mo
Opacity	NDA/closed-door pricing, “disguised” in partnerships	“Rates vary by orders of magnitude” (Quartz)
Exclusion	Worthwhile only for large premium sellers	~52 deals; 38 sellers; individuals earn “a few dollars”
No fit measure	Buyers can't tell if data meets their need	Marketplaces solve “search” only (Data Boutique)
No price discovery	No bidding; valuations are guesses	Kodak 5% vs Nortel 450% of estimate
Unclear rights	License omitted/mislabeled; resale unclear	65% of HF licenses omitted/mislabeled
Illiquidity	Marketplaces <2% of volume; dead code unsold	Most failed-startup code never monetized

3. The Bitcode model — a measured market for code IP

3.1 AssetPacks: the unit

An AssetPack (AP) is a **measured patchfile** — a synthesized unit of code IP, not a raw file dump. Depositing and reading are symmetric: in both, an AP is synthesized and measured. The difference is direction — deposit synthesizes from your (partially Obfuscated) repository; read synthesizes from the pool of relevant deposited APs (to satisfy stated Needs).

Step	Deposit (supply)	Read (demand)
1	Connect a repository	Connect a repository
2	Mark obfuscations — what to exclude from synthesis	Specify read needs — what the AP(s) must satisfy
3	AP synthesized from your obfuscated repo	AP synthesized from relevant deposited APs
4	Absolute measurements: static analysis, AI evaluation, scores	Absolute measurements + dynamic, need-relative scores
5	AP is listed and tradeable in the live market	Review APs with presented fit scores, via Bitcode AI
6	Earn, BTD (1/2)+BTC by measured contribution when packs are bought	Receive the AP (unlimited rights) + minted $BTD(1/2)

Obfuscation is what makes deposit safe: a holder can expose non-moat IP while withholding secrets, so legacy and proprietary repositories become sellable without giving away the crown jewels.

3.2 Measurement: quality × quantity

Measurement is the core primitive — it is what lets code be “measured, commoditized, bought, and sold.” Every pack carries **absolute** measurements (quality, established by exhaustive static analysis and AI evaluation) and, on a read, **dynamic** measurements that score how strongly the pack satisfies the buyer's stated need. The weighted sum of a purchased pack's measurements yields a single volumetric scalar — its *quantity of knowledge*. In short: **measurements demonstrate quality; $BTD demonstrates quantity**; price is set by live bidding and is heavily tied to both.

3.3 $BTD: the asset

$BTD is a fungible, deflationary digital asset backed by purchased AssetPacks. On purchase, the pack's knowledge scalar determines how much $BTD is **minted and delivered to the buyer+seller (50:50 split) along with the AP**. It plays three roles at once: an **in-platform currency** (medium of exchange), a **proof/receipt of knowledge** acquired, and a **tradeable, speculative asset**. Because supply is knowledge-gated and deflationary, buying is simultaneously consumption and value accrual — every purchase both delivers IP and mints a claim on the market's growing stock of knowledge.

3.4 Rights, pricing, and the two-sided loop

Depositing grants any buyer **unlimited rights** to the IP once purchased — they may use, resell, hoard, or open-source it. For simple economic reasons, **reselling and hoarding** are the most common outcomes: APs behave like assets, not just inputs. Pricing is continuous and competitive (bidding, real-time trading), so — unlike closed-door channels — the market price is visible and comparable. The result is a self-reinforcing loop: more deposits deepen the pool synthesis draws from; better measurement improves buyer confidence; liquidity and $BTD reward both sides for participating.

4. Supply side — who deposits, and why

4.1 The deposit flow & supplier economics

Depositing is a few clicks: connect a repo, mark obfuscations, receive synthesized-and-measured APs to review before listing. Suppliers are paid **by their measured contribution** to every purchased pack — including read-synthesized packs assembled from many deposits. This is the decisive break from today: there is no negotiation, no minimum deal size, no NDA, and no need to be a premium rights-holder. A holder of fragmentary or long-tail code earns whenever that code measurably helps satisfy a buyer's need.

Why suppliers win here. Fastest path from IP to visible-and-sold (clicks, not a ~9-month negotiation); transparent, bid-set prices instead of closed-door quotes; obfuscation lets sensitive repos be sold safely; and the long tail — legacy code, dead code, solo-hacker work — is monetizable for the first time.

4.2 The full supplier universe

Supply spans three pools from the market analysis — positive-ROI recycled data, dead code, and IP made to be sold — expanded into concrete depositors:

Supplier	What they deposit	Why they deposit
Enterprises (banks, telco, retail, mfg, healthcare)	Legacy, non-moat, deprecated internal code	Monetize sunk cost; obfuscation keeps moats private
Software vendors	End-of-life products and old versions	Recycle IP that no longer sells commercially
Defunct / pivoted startups (founders, estates, assignees, investors)	Whole abandoned codebases ("dead code")	Recover value that bankruptcy otherwise destroys
Frontier solo-hackers & independent devs	Novel implementations, kernels, agents	Research-for-cash without needing to be hired
Open-source maintainers & foundations	Permissive / dual-licensed code	Fund continued development
Agencies, consultancies, dev shops	Reusable modules, boilerplate, patterns	Monetize repeat work across clients
Universities & research labs	Reference & research implementations	Tech transfer; research-for-cash
ML / data teams	Eval suites, RL environments, data-gen scaffolds	High-value, scarce, hard-to-synthesize IP
Model-tuning companies	Domain-specific code & tuning recipes	Monetize vertical IP between client engagements
Model-building agents (autonomous)	Code they generate	Agent-to-market monetization at machine speed

5. Demand side — who reads and buys, and why

A buyer connects a repo or states a need, reviews synthesized APs carrying dynamic fit scores, and buys in a couple of clicks — receiving the AP with unlimited rights plus minted $BTD. Two distinct engines drive demand.

5.1 Two demand engines: consumption & speculation

- **Consumption (utility).** Model builders and tuners buy APs that measurably satisfy a capability need — rare domains, high-quality implementations, RL environments — with the fit scored before purchase. The value is the IP itself, used to train or tune.
- **Speculation & holding (financial).** Because rights are unlimited and $BTD is deflationary and knowledge-backed, buyers also acquire APs to resell or hoard. High-knowledge packs are assets; scarcity is visible; and every purchase mints $BTD. This financial engine deepens liquidity that pure-utility marketplaces never achieve.

Why buyers win here. Need-fit is measured before you pay (no incumbent does this); one open market replaces per-vendor sales calls and trials; acquisition is instant; and every purchase also mints a knowledge-backed asset ($BTD), so buying compounds rather than just spends.

5.2 The full buyer universe

Buyer	Primary motive	What they seek
Frontier labs (OpenAI, Anthropic, Google, Meta, Amazon, Microsoft, xAI, Mistral, Cohere, DeepSeek, Alibaba)	Consumption — training	Rare, high-quality, rights-clean code; scarce domains; RL environments
Other model builders (NVIDIA, Databricks, Reka, Contextual, national/vertical)	Consumption — training	Domain breadth without first-party data moats
Model-tuning / fine-tuning firms (Together, Fireworks, Predibase, Snorkel, Lamini)	Consumption — customization	Vertical & domain code for client models
Enterprises building internal models	Consumption — customization	Industry-specific code and patterns
Managed data vendors (Scale, Surge, Mercor)	Resupply	APs to curate and repackage for labs
Model-building agents (autonomous)	Consumption — at machine speed	APs that satisfy a build-need on demand
Traders, funds, market-makers	Speculation / liquidity	Underpriced high-knowledge APs; $BTD exposure
Strategic hoarders (labs, enterprises)	Denial / scarcity	Scarce domain APs to keep from rivals
Open-source foundations & archivists	Liberation	APs to purchase and open-source
Compliance-driven buyers (regulated industries)	Provenance / audit	Rights-cleared, provenance-tracked APs (EU AI Act)

6. How Bitcode is differentiated

Three pillars define why Bitcode beats data-brokers — each directly attacking a friction from §2.

- **1. Fastest to visible, bought, and sold.** Getting IP onto the market and transacted is a matter of clicks and a live bid — not a multi-month, NDA-bound negotiation that may collapse. Time-to-liquidity collapses from ~9 months to minutes.
- **2. Most live, real-time market.** Prices are set in the open by bidding and real-time trading — not behind closed doors. Real-time markets continuously price AssetPacks and $BTD, so value is visible and comparable, ending the guesswork that made the same IP clear at 5% or 450% of estimate.
- **3. Easiest experience, full-stop.** A couple of clicks to receive AssetPacks to review — to buy, or to sell. No sales calls, no per-vendor trials, no bespoke contracts. Review-to-buy and review-to-sell are the whole workflow.

Underlying enablers make the three pillars possible: **measurement** (objective quality + a quantity scalar, so IP can be priced and compared); **synthesis + obfuscation** (safe deposit and need-shaped reads); **clean, unlimited rights** (resale and hoarding, so IP is an asset); and **tokenization via $BTD** (liquidity and speculative depth).

6.1 Competitive positioning — the white space

The tokenized data/IP field has a measurement camp and a provenance camp, and no one bridges them for code with live pricing and clean resale. Bitcode occupies that intersection.

Player	Measures quality?	Live price discovery?	Rights / notes
Bitcode	Yes — absolute + dynamic	Yes — live bidding	Unlimited resale; native to code IP
Story / DATA Foundation	No (provenance only)	No (mint fee + royalty)	Royalty-encumbered; general IP/data
Ocean Protocol	Retired (2024)	Partial (AMM)	Access licenses; not code
Vana	Weak (staking)	No	DataDAO tokens; not code
Sahara AI	Weak (attribution)	No	Non-transferable revenue shares
Numerai / Bittensor	Strong (stake / validators)	No (closed system)	Predictions / model outputs, not IP
Scale / Surge / Mercor	Internal QA only	No (bespoke)	Work-for-hire; manufactured data
AWS / Snowflake / HF	No	No	Listing only; rights often unclear
IPXI / IPwe	No	No	Failed IP exchanges (2015 / 2024)

7. Where Bitcode's niche is

Bitcode's niche is the intersection no incumbent occupies: **measured, rights-clean, liquid, tokenized code IP**. Concretely, it wins where today's channels are weakest:

- **The long tail of code.** Legacy and non-moat repositories, solo-hacker work, and fragmentary IP — too small for a licensing deal, too specific for a marketplace listing — become monetizable via measured contribution.
- **Dead code.** Codebases trapped in defunct/pivoted companies, which today clear at ~5% of value after ~11 months, *become instantly visible and price-discovered*.
- **Need-shaped acquisition.** Buyers who need a specific capability get a synthesized pack with measured fit — not a dataset they must evaluate blind.
- **Financialized knowledge.** $BTD makes code IP a tradeable asset class, attracting liquidity (traders, hoarders) that pure-utility venues cannot.
- **Compliance-grade code.** Rights-cleared, provenance-tracked packs satisfy audit requirements (EU AI Act) that scraped corpora and the GitHub-Copilot / Anthropic-settlement risk cannot.

8. What real transactions look like on Bitcode

Bitcode is more optimal than a broker for a specific class of deals — small, fast, measured, long-tail, or asset-like. Six representative archetypes:

Archetype 1 — Enterprise legacy-code monetization

Setup. A bank has millions of lines of retired core-banking code — valuable patterns, but wrapped around secrets it can't expose.

On Bitcode. It connects the repo, obfuscates PII, security, and proprietary logic, and Bitcode synthesizes measured APs. Many buyers read against them; the bank earns by measured contribution on every purchase.

Why it's optimal here. No broker would assemble a bespoke deal for fragmentary internal code, and no buyer would negotiate for it. Bitcode turns dormant, non-moat code into a recurring revenue stream in days.

Archetype 2 — Dead-code liquidation

Setup. A defunct robotics startup's assignee holds a rich codebase that would otherwise sell — if ever — as patents in bankruptcy court.

On Bitcode. The whole codebase is deposited; APs are synthesized, measured, and listed. Buyers bid; a transparent clearing price emerges within clicks.

Why it's optimal here. It replaces an ~11-month, monopsony-driven process that clears at ~5% of estimate with instant, live price discovery — value recovered instead of forgotten.

Archetype 3 — Need-driven acquisition by a model builder

Setup. A lab needs a scarce capability — say, high-assurance embedded-firmware patterns or rare-language concurrency code.

On Bitcode. It states the need; Bitcode synthesizes an AP from relevant deposits and attaches dynamic scores showing exactly how well it satisfies the need. One click to buy; it receives the AP plus minted $BTD.

Why it's optimal here. Versus months of sourcing with no fit guarantee, the buyer gets measured fit in minutes — and the purchase also mints a knowledge-backed asset.

Archetype 4 — Speculative resale & hoarding

Setup. A trader sees an underpriced pack with a high measured knowledge scalar.

On Bitcode. They buy it outright (unlimited rights), then hold or resell as scarcity and $BTD value accrue.

Why it's optimal here. No incumbent enables this at all — code IP as a tradeable asset is unique to Bitcode's rights model and $BTD, and this liquidity underwrites the whole market.

Archetype 5 — Solo-hacker research-for-cash

Setup. An independent researcher writes a novel implementation (e.g., a new attention kernel) with no path to sell it short of getting hired.

On Bitcode. They deposit it; exhaustive measurement scores its quality; buyers bid; they are paid by measured contribution.

Why it's optimal here. The long tail of frontier talent monetizes directly — no employment, no negotiation, no gatekeeper.

Archetype 6 — Compliance-grade sourcing

Setup. A regulated enterprise must document training-data provenance (EU AI Act) and cannot risk scraped, unlicensed code.

On Bitcode. It buys rights-cleared, provenance-tracked APs whose measurements and rights trail satisfy audit out of the box.

Why it's optimal here. Scraped corpora carry the exact risk that produced the GitHub-Copilot litigation and Anthropic's $1.5B settlement; Bitcode's cleared packs remove it.

9. Risks & open questions

Risk / question	Why it matters
Measurement trust	The whole market rests on measurements being objective and gameable-resistant; adversarial deposits must not inflate scores. (SWE-bench-style test-pass scoring is a strong, verifiable anchor.)
Rights integrity at deposit	Unlimited-rights resale requires airtight proof the depositor had the right to deposit; provenance and warranties must be enforceable.
Liquidity cold-start	Two-sided markets need both sides at once; IPXI and IPwe died on thin demand. Seeding buyers (or $BTD incentives) early is critical.
Synthesis leakage	Obfuscation must guarantee withheld IP can't be reconstructed from synthesized or read-assembled packs.
$BTD reflexivity	A knowledge-backed, speculative token can decouple from IP utility; token design must keep price tied to real measured value.
Buyer vertical integration	As with Meta–Scale, large buyers may try to internalize supply; differentiated, exclusive-feeling packs and network effects are the defense.

10. Sources

Friction and competitor facts were gathered via multi-source web research and cross-checked; product mechanics reflect Bitcode's stated design. Key sources:

- Quartz — The price of AI training data ($5M–$250M): qz.com/ai-training-data-pricing-licensing-deals-market-052126
- Press Gazette — publisher AI deals & lawsuits (OpenAI–NYT ~9 months): pressgazette.co.uk/platforms/news-publisher-ai-deals-lawsuits-openai-google

- Neudata — AI data-licensing market analysis (52 deals, 38 sellers): neudata.co/blog/ai-data-licensing-market-analysis
- Data Boutique — Generic data marketplaces are broken (<2% of volume): blog.databoutique.com/p/generic-data-marketplaces-are-broken
- arXiv 2502.04484 — empirical analysis of Hugging Face dataset licensing (65% omitted/mislabeled)
- Marketplace / APM — the only profitable AI companies supply training data (2025)
- IEEE Spectrum — The Lowballing of Kodak's Patent Portfolio (2014)
- CoinDesk — Vana data-backed token standard (VRC-20, 2025)
- Messari / Story Documentation — Story Protocol & DATA Foundation "Trace" (2025–26)
- Numerai / Chainwire — staking, burn, $500M valuation (2025)
- Contreras, "FRAND Market Failure" — IPXI post-mortem (2016); IAM — IPwe bankruptcy (2024)
- SWE-bench / arXiv 2509.16941 — patch-generation benchmark & measurement context
- Companion: Bitcode Market Analysis — Research Brief (sizing, accounts, take-rate comps)